



Dynamics and Control R&D for Energy Systems

Energy Systems Initiative

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Office of Research & Development

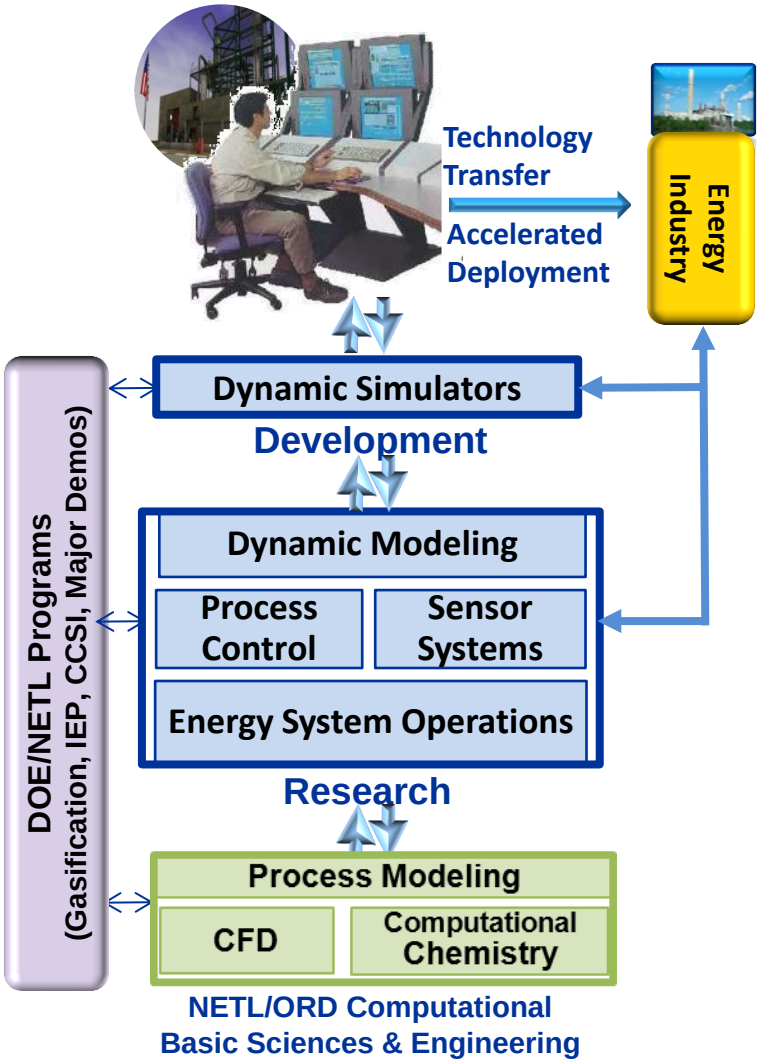
March 9, 2014

U.S. DEPARTMENT OF ENERGY | National Energy Technology Laboratory

Overview

Mission: Operational Excellence for Energy Systems

- **Goals and Objectives**
  - Develop and deploy state-of-the-art dynamic modeling, simulation, and control tools
  - Leverage in R&D to maximize efficiency and profitability of energy system operations, while reducing negative environmental impact and improving safety
- **Dynamic Simulators Development**
  - High-fidelity real-time dynamic simulators
  - Full-scope operator training systems (OTSs)
  - 3D virtual immersive training systems (ITSs)
- **Advanced Research**
  - Dynamic Modeling
  - Advanced Process Control
  - Sensor Systems
  - Energy System Operations



Dynamic Simulator Development

Energy System Applications

- **Integrated Gasification Combined Cycle (IGCC) with CO_2 Capture [OTS/ITS]**
- **Natural Gas Combined Cycle (NGCC) [OTS]**
- **Supercritical Once-Through (SCOT) Pulverized Coal [OTS]**
- **Subcritical Pulverized Coal (SubPC) [OTS]**
- **Oxy-Coal Carbon Capture (OCCC) [OTS]**
- **Shale Gas Processing (SGP) [OTS]**









Key:

- Deployed (Completed Phases 1-5)
- Under Development (Completed Phases 1-2)
- Scoping (Completed Phase 1)

Dynamic Simulator Development

IGCC with CO_2 Capture – OTS/ITS

- **Motivation**
  - Flexible technology for clean power generation
- **Deployed OTS at NETL and WVU (2011)**
  - Software: DYNSIM v 4.5.3 / InTouch v9
- **Deployed ITS at NETL and WVU (2012)**
  - Software: EYESIM v1
- **Collaborated with Development Partners**

NETL-RUA Regional University Alliance · West Virginia University · URS · invensys · FCS Fossil Consulting Services, Inc. · EPRI Electric Power Research Institute

- **Collaborated with Industry**

AEP · bp · DOOSAN · SOUTHERN COMPANY · GREAT RIVER ENERGY

- **Distributed Run-Time OTSs for Internal Use**
  - EPRI, BP, Doosan, Southern Company
- **New Developments in Progress**
  - SW Copyright License Agreement with Invensys
  - ITS upgrade to EYESIM v2.0









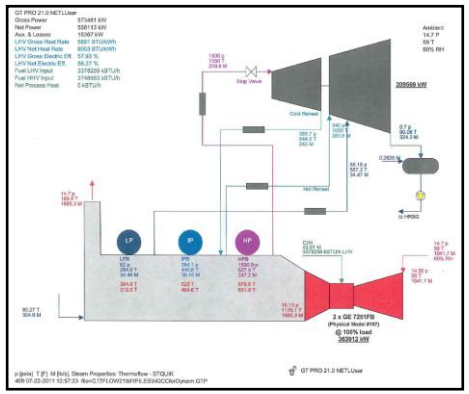
Zitney, S.E., E.A. Liese, P. Mahapatra, R. Turton, D. Bhattacharyya, and G. Provost, "AVESTAR Center: Dynamic Simulation-based Collaboration Toward Achieving Operational Excellence for IGCC Plants with Carbon Capture," Proc. of 29th Annual Int'l Pittsburgh Coal Conf., Pittsburgh, PA, Oct. 15-18 (2012).

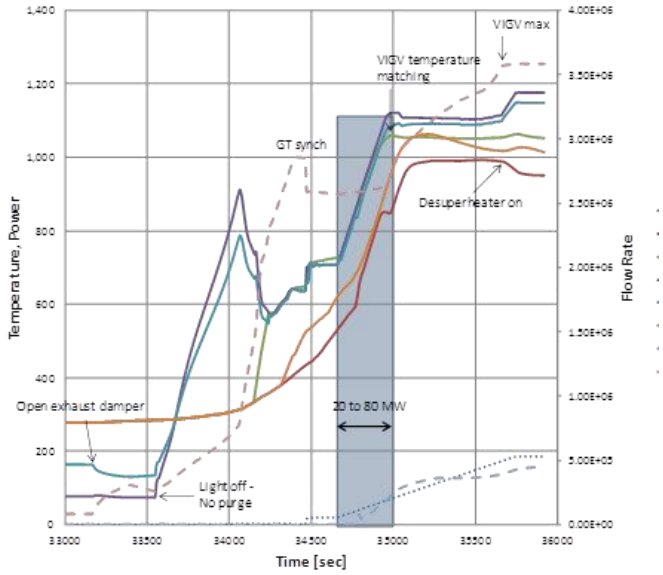
Dynamic Simulator Development

Natural Gas Combined Cycle (NGCC) – OTS

- **Motivation**
  - Shale gas plays leading to low gas prices
  - Stricter regulation for coal plants
  - Integration of growing amounts of renewable power
- **Completed Steady-State Design**
  - 2-on-1 design with 574 MW gross power
  - Two GTs (182MW each) x One ST (210MW)
  - Two 3-pressure HRSGs (1890, 385, and 62 psia)
- **Completed Prototype Dynamic Model**
  - Leveraged CC portion of IGCC dynamic simulator
  - Modified HRSG heat exchangers and drums
  - Modified steam turbine to match new conditions
  - Achieved stable full-load and tested ramping
- **New Developments in Progress**
  - Update process controls and HMIs
  - Collaborate with Invensys under CRADA to complete development, testing, and deployment
  - Collaborate on cycling studies with NRECA under CRADA (Associated Electric Coop, NGCC Power Plant, Dell, AR)
- **Potential Future Work**
  - Add hooks for post-combustion CO_2 capture (NG-CCS)
  - Integrate with variable renewable generators

NGCC Plant Design in GT-PRO





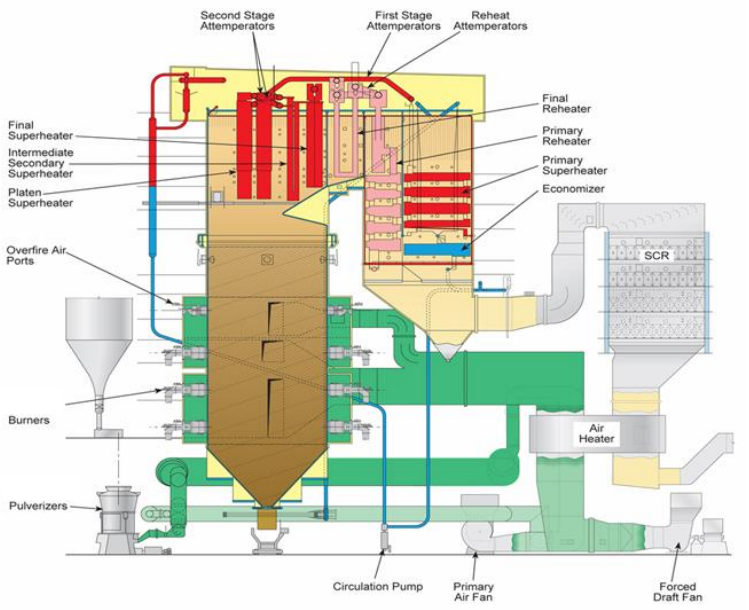
Startup of one GT after hot-hold for NGCC (Liese and Zitney, 2013)

Liese, E.A. and S.E. Zitney, "A Dynamic Process Model of a Natural Gas Combined Cycle – Model Development with Startup and Shutdown Simulations," Proc. of ASME 2013 Power Conference, Boston, MA, July 29 – August 1 (2013).

Dynamic Simulator Development

Supercritical Once-Through (SCOT) Pulverized Coal - OTS

- **Motivation**
  - Post-combustion CO_2 capture and cycling
- **Accomplishments/Results**
  - Generated functional design specification
    - Once-through water/steam circulation system
    - Enables quicker startup, variable pressure load-following, and on-off cycling
  - Developed DYNSIM dynamic model
  - Prototyped process controls and InTouch HMIs
- **New Developments in Progress**
  - Collaborate with Invensys (CRADA) to complete:
    - Development of HMIs, and operating procedures
    - Testing and deployment
- **Potential Future Work**
  - Add air quality control systems (AQCS)
    - Baghouse (PM), SCR (NOx), FGD (SO_2)
  - Implement process/heat hooks for CO_2 capture
  - Integrate with CO_2 capture dynamic models
  - Conduct transient studies, including startup, shutdown, cycling, load-following, and variable CO_2 capture rates

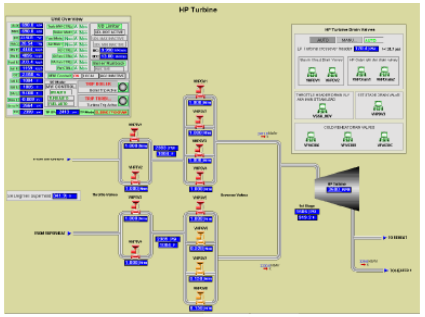




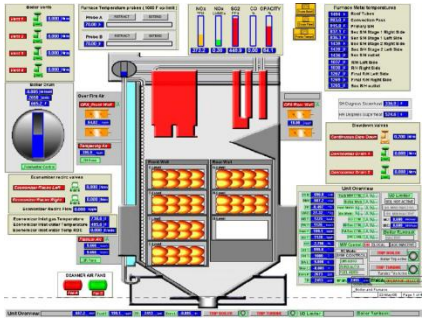
Wilbers, D. and S.E. Zitney, "Invensys and DOE Partnering to Create Two Generic Simulators for Carbon Capture Studies," 56th Annual ISA Power Industry Division (POWID) Symposium, Orlando, FL, June 2–7 (2013).

Dynamic Simulator Development
Subcritical Pulverized Coal (SubPC) - OTS

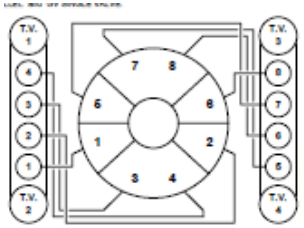
- **Motivation**
  - Post-combustion CO_2 capture and cycling
- **"Generic" but detailed design**
  - Sidewall-fired drum boiler
  - Westinghouse steam turbine (690 MW)
  - 2413 psig throttle pressure
  - 1005°F main and reheat steam temperature
  - Lube oil, steam seal, etc.
  - Full control system, permissives, alarms
  - Cold-metal start
- **DYNSIM Power for dynamic model and HMI**
- **Training literature developed by FCS**
- **Accomplishments/Results**
  - Reviewed features, capabilities, and documentation for generic SubPC dynamic simulator/OTS (Invensys)
- **Potential future work**
  - SubPC cycling project(s)



DYNSIM Process Model



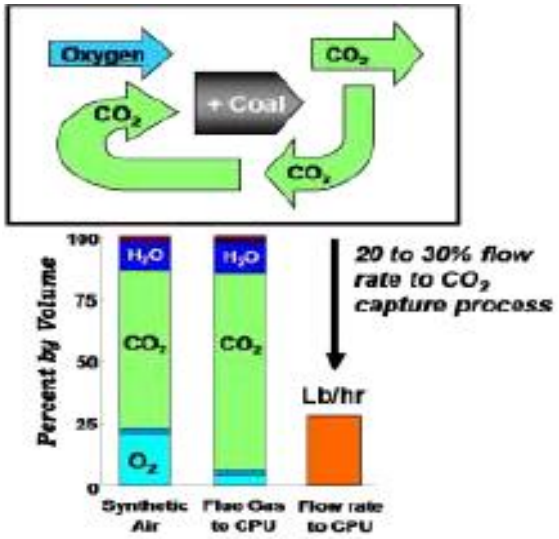
DYNSIM HMI



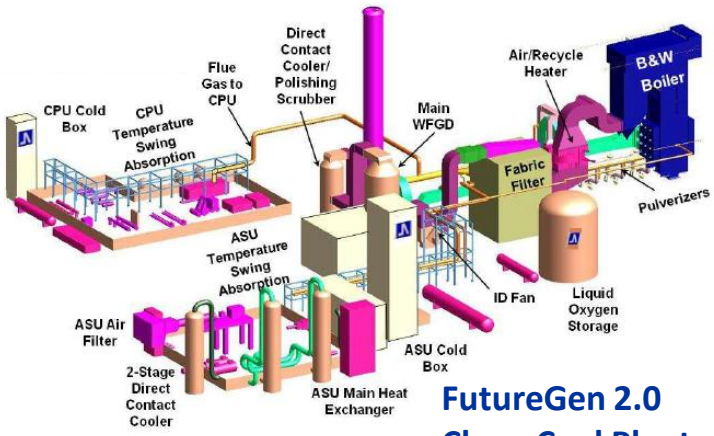
Steam Turbine Configuration

Dynamic Simulator Development
Oxy-Combustion Clean Coal (OCCC) – OTS/ITS

- **Motivation**
  - Nitrogen removed from process
  - Flue gas contains mainly CO_2 and H_2O
  - Compression and CO_2 purification unit (CPU)
- **FutureGen 2.0** FutureGen™
  - Meredosia Power Station, IL
  - 200 MWe, Unit 4, 160Bar/540C/540C
  - Babcock & Wilcox (B&W)
    - Purpose-built oxy-PC boiler
  - Air Liquide (AL)
    - Air separation unit (ASU) and CPU
- **Accomplishments/Results**
  - OTS/ITS(Ph1): Generated proposal for FG 2.0 Phase II – FEED
- **New Developments in Progress**
  - Participate in NETL collaboration with B&W and AL on FEED
- **Potential Future Work**
  - Develop and deploy OCCC OTS/ITS
  - Use FG 2.0 as template for NETL Large-Scale Clean Coal and Power System Demonstrations



Oxy-Coal Combustion

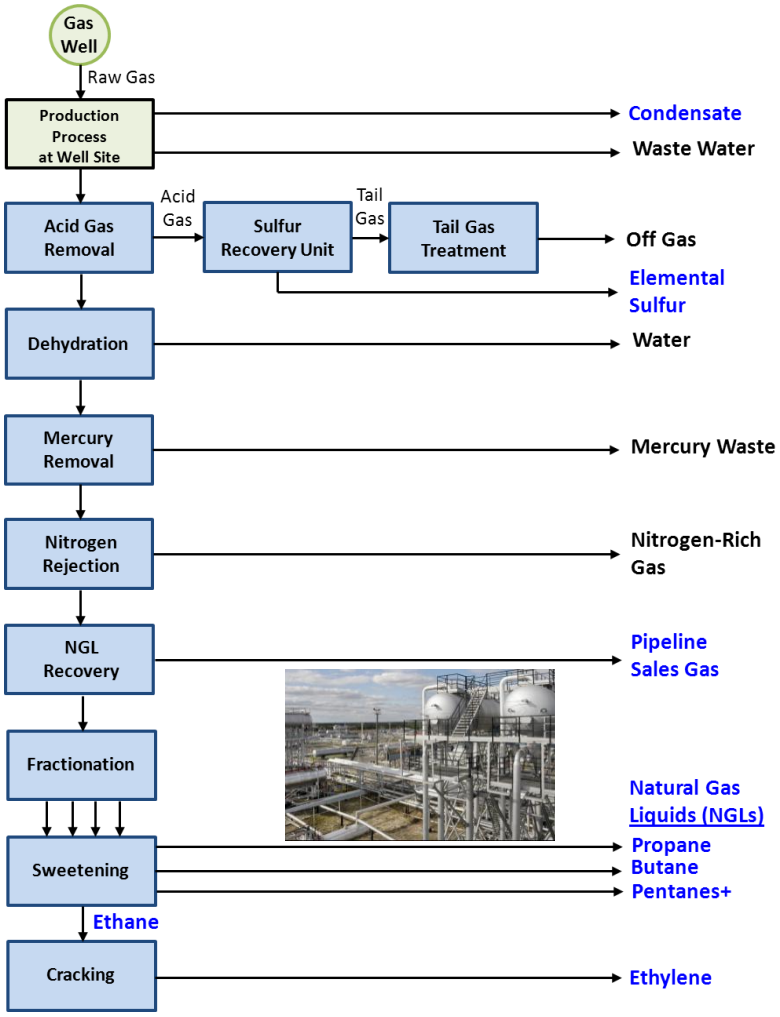


FutureGen 2.0 Clean Coal Plant

Dynamic Simulator Development

Shale Gas Processing (SGP) – OTS/ITS

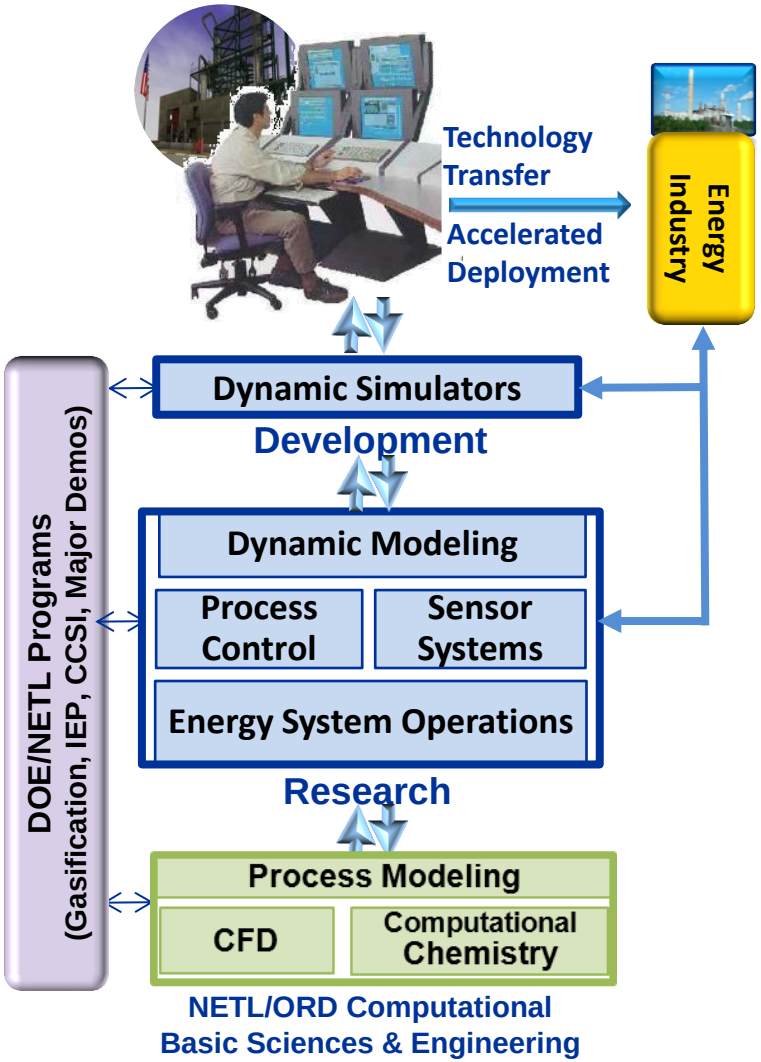
- **Motivation**
  - Regional Marcellus/Utica shale gas plays
  - Large, highly-integrated, multi-purpose facilities with high-value assets
  - Efficient operation and control of treatment and separation processes to generate pipeline-quality natural gas and natural gas liquids (NGLs)
- **Potential Partners**
  - **ShaleNET**
    - Industry Workforce Training; Drilling/fracking to SGP
    - 20 Approved Training Providers
      - ✓ Westmoreland County Community College, PA
      - ✓ Pierpont Community & Technical College, WV
- **Accomplishments/Results**
  - Completed scoping study for SGP OTS/ITS
  - Discussed collaboration opportunities with ShaleNET industry partners
- **Potential Future Work**
  - Collaborate with ShaleNET and industry partners on SGP OTS proposal to US. Dept. of Labor



Overview

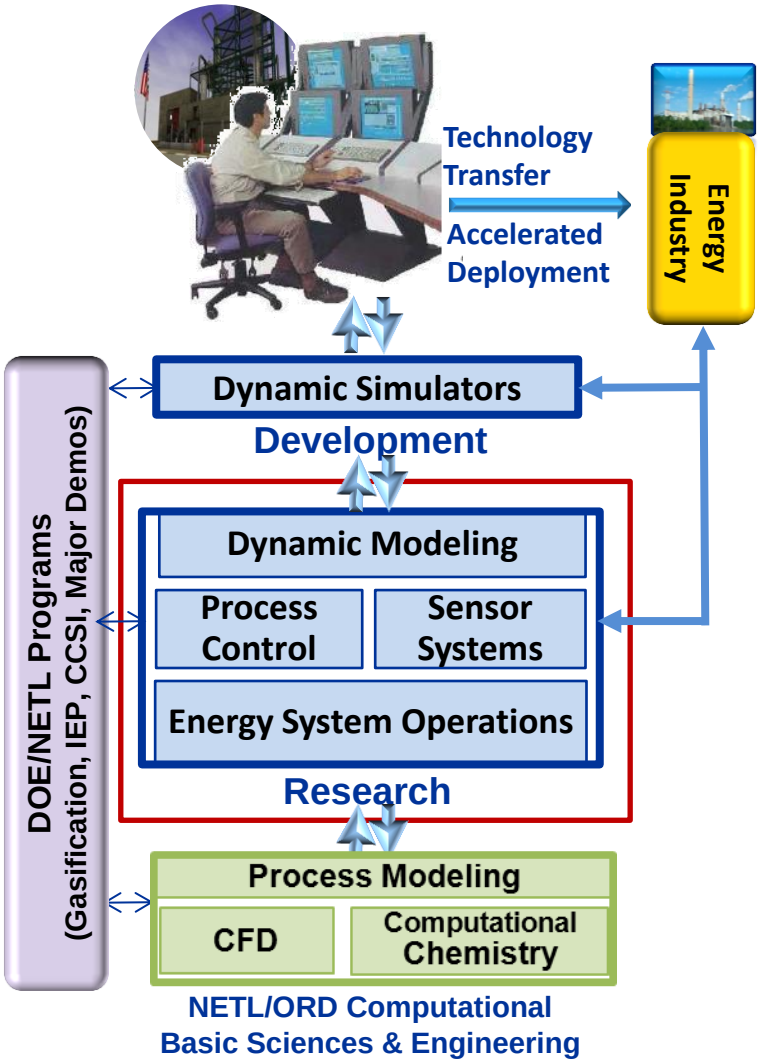
Operational Excellence for Energy Systems

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  - Advanced Process Control
  - Sensor Systems
  - Energy System Operations



Advanced Research Program
Dynamics and Control

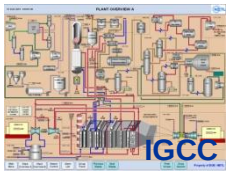
- **Dynamic Modeling**
  - Plantwide/Process/Equipment
  - Physical and Chemical Submodels
  - Dynamic reduced models (D-RMs)
- **Process Control**
  - Plantwide Control System Design
  - Advanced Process Control
- **Sensor Systems**
  - State estimation, Disturbance rejection
  - Process monitoring, Fault diagnosis
- **Energy System Operations**
  - Startup, shutdown
  - Ramping, cycling, load following
  - Safety and environmental analysis



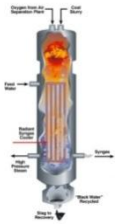
Dynamic Modeling

Plantwide/Process/Equipment

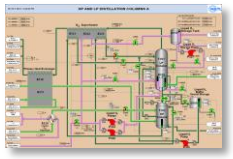
- **Plantwide**
  - IGCC [APD, DYNSIM]
  - NGCC [DYNSIM]
  - SCOT [DYNSIM]



- **Process/Equipment**
  - Entrained-Flow Gasifier [ACM]
    - Approach: 1-D PDE [ACM]
      - Sub-models
        - Reaction Kinetics
        - Recirculation
        - Slag Flow/Penetration

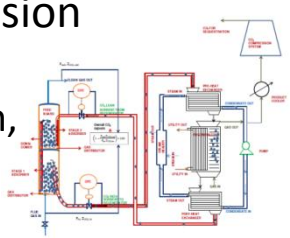


    - Approach: Multizonal [DYNSIM]
  - Air Separation Unit (ASU)
    - Low/Elevated-Pressure Cryogenic ASUs [APD, DYNSIM]



  - Sulfur Capture
    - Claus unit and reactor [APD]
      - Sub-models
        - Reaction Kinetics

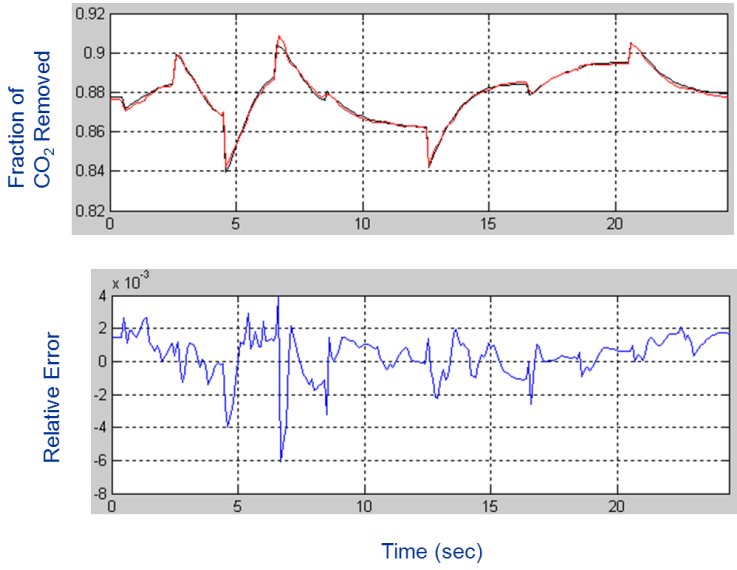
- **Process/Equipment**
  - CO_2 Capture/Compression
    - Post: Solid Sorbent Capture, Regeneration, and Transport [ACM]
      - Approach: 1-D PDE



    - Post: Liquid Solvents Capture/Regen [ACM, gPROMS]
      - Approach: Rate-Based [gPROMS]
    - Pre: Selexol [APD, DYNSIM]
    - Pre/Post: Multi-stage CO_2 Compression [DYNSIM, APD/ACM]
  - Steam Turbines [ACM, DYNSIM]
- **Commercial Software**
  - Aspen Plus Dynamics [APD], Aspen Custom Modeler [ACM] (AspenTech)
  - DYNSIM (Invensys)
  - gPROMS (PSE)
  - MATLAB/Simulink (MathWorks)

Dynamic Modeling
Dynamic Reduced Models (D-RMs)

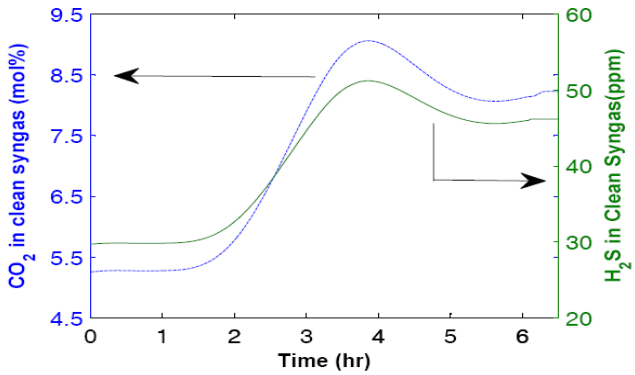
- **CCSI D-RM Builder**
  - **Data-driven Black-Box Methods**
    - Nonlinear Auto-Regressive Moving Average (NARMA) based on Neural Networks
    - Decoupled A-B Net (DABNet)
      - Sentoni, Biegler, Guiver, Zhao, *AIChEJ* (1998)
  - Automatic D-RM Generation
    - Run high-fidelity "ACM" models in Simulink
    - Create data-driven D-RMs as MATLAB models
  - GUI Driven Workflow
    - Configuration → Space-filling (LHS) → Training (Ramp Changes)→ Post-processing → Export
  - Test Cases
    - VandeVusse Reactor (input-multiplicity)
    - CSTR (multiple SS, unsteady operation)
    - 4 Tank (multiple transmission zeroes)
    - 1D PDE Dynamic BFB CO_2 Capture Adsorber
  - Product Release
    - D-RM Builder R1 – Oct. 2013
    - D-RM Builder R2 – Oct. 2014
      - Enhanced D-RM Error Estimation
      - Training algorithm to prevent D-RM Overfitting
      - UQ-enabled D-RM with uncertainty (covariance) matrices and likelihood to assess goodness of fit
  - **Order Reduction Methods**
    - Spatial Reduction
      - Proper Orthogonal Decomposition (POD)
    - Temporal Reduction
      - Eigenvalue Analysis and Quasi-steady State Approximation for Fast States
    - Test Case
      - 1D PDE Dynamic BFB CO_2 Capture Adsorber



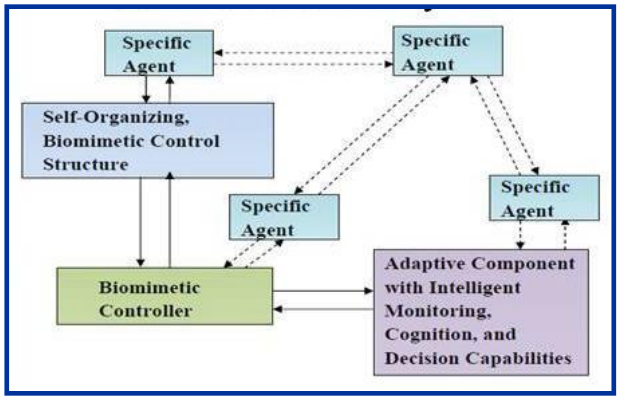
D-RM (DABNet) vs. ACM Output Data for BFB

Process Control

- **Plant-wide Control System Design for Advanced Energy Systems**
  - WVU, Prof. Bhattacharyya, Jones
- **Advanced Process Control for IGCC Power Plants with CO_2 Capture**
  - WVU, Prof. Bhattacharyya; NETL, Dr. Mahapatra
- **Advanced Process Control Framework for Carbon Capture System Operations**
  - MPC, State Estimation, and UQ
  - CCSI: NETL, Drs. Mahapatra, Zitney; WVU, Prof. Bhattacharyya; LLNL, Dr. Ng
- **Integrated Biomimetic Control Framework for Advanced Energy Plants**
  - Self-Organization, Distributed Intelligence, Adaptability, Intelligent Monitoring, Cognition, and Decision Capabilities
  - WVU, Profs. Bhattacharyya, Lima, Turton, Perhinschi; VRI, Dr. Diwekar



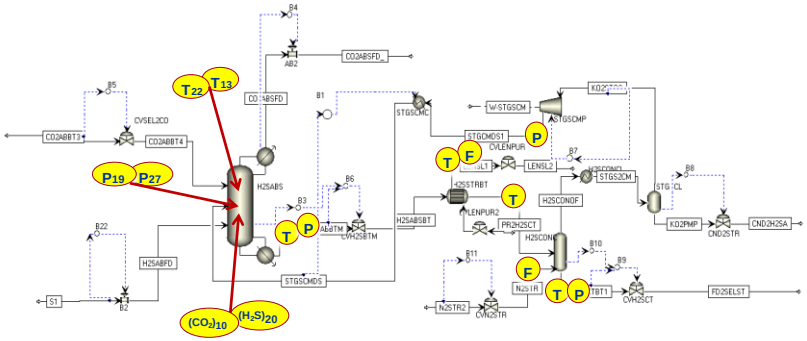
IGCC with CO_2 Capture response to ramp increase in coal feed flowrate (Bhattacharyya et al., 2011)

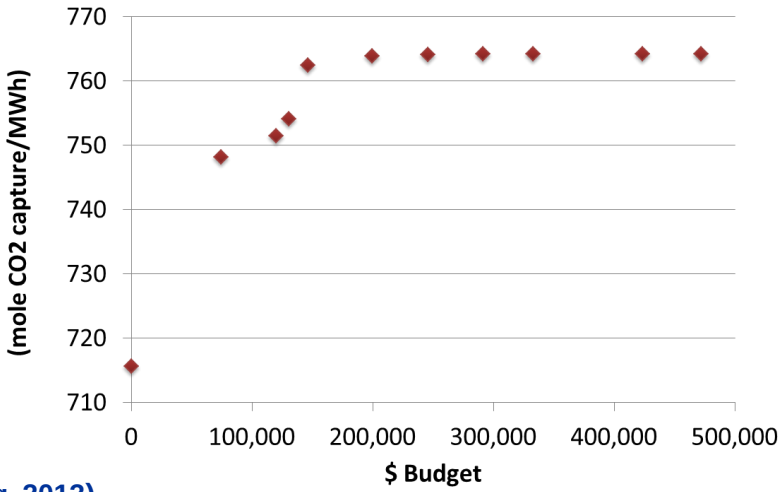


Schematic of Proposed Biomimetic Control Framework (Bhattacharyya et al., 2014)

Sensor Systems

- **Development of Optimal Sensor Placement Algorithms for Energy Systems**
  - Application: IGCC Power Plant with CO_2 Capture
  - Objective: Maximize Plant Efficiency
  - WVU, Prof. Bhattacharyya, Prof. Turton, Paul



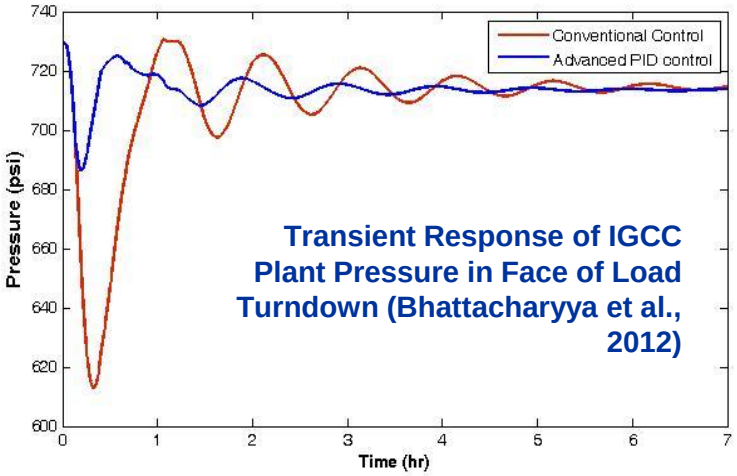


(Paul, Bhattacharyya, Turton, Zitney, AIChE Annual Meeting, 2013)

- **Multi-Objective Optimal Sensor Deployment Under Uncertainty for Advanced Power Systems**
  - VRI, Dr. Diwekar; WVU, Prof. Bhattacharyya
- **Smart Refractory Sensor Systems for Wireless Monitoring of Temperature, Health, and Degradation of Slagging Gasifiers**
  - WVU, Profs. Sabolsky, Bhattacharyya, Graham, Kutathumani; ANH Refractories, Palmisiani

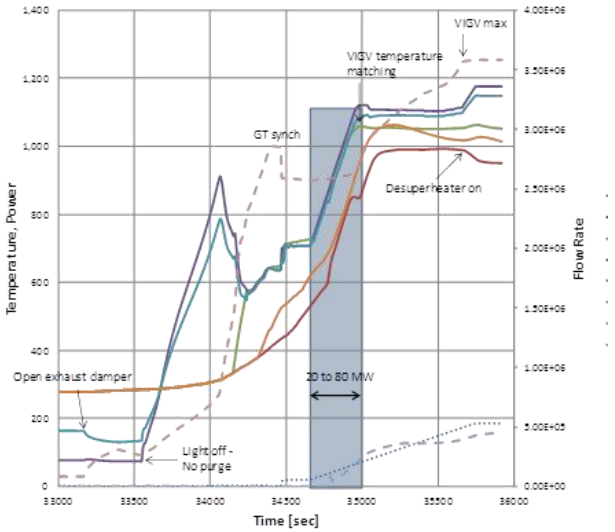
Energy System Operations

- **IGCC Power Plant Load Following**
  - Improved GT-Lead/Gasifier-Follow and Gasifier-Lead/GT-Follow modes
  - Improved ramp rates
  - WVU, Prof. Bhattacharyya
  - NETL, Mahapatra



Transient Response of IGCC Plant Pressure in Face of Load Turndown (Bhattacharyya et al., 2012)

- **NGCC Shutdown, Startup, and Cycling**
  - Operational strategies to reduce plant derates, emissions, profit loss, and equipment damage
  - CRADA with the National Rural Electric Cooperative Association (NRECA)
  - NETL, Liese, Zitney



NGCC Shutdown of one GT from full-load conditions (Liese and Zitney, 2013)

Thank you! Questions?



For more information, please contact:
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